

جامعة الأزهر - غزة
Al Azhar University - Gaza



**FACULTY OF ENGINEERING AND INFORMATION
TECHNOLOGY**

Practical Training (1)

**Final report of the practical training at Al-Shifa
hospital**

prepared by

Mahmoud S Al-Yazji

20201333

SUPERVISED BY

Dr-Abdullah Al-Sayed

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Introduction

Dr. Abdullah Alsayed, the head of the mechatronics and medical devices engineering department at Al-Azhar University,

I hope this report finds you well.

I am thrilled to present my final academic report, which provides a comprehensive overview of my practical training journey at Al-Shifa Hospital's Medical Engineering Department. Under the direct guidance of Eng. Abdulhamid Syam, the department's head of medical devices, I engaged in a month-long immersive field training, accumulating a total of 60 hours. Throughout this period, I rotated through crucial departments including dialysis, maternity, ESWL unit, and the central maintenance workshop. The primary objective of this training was to translate theoretical knowledge into practical solutions for real-world medical engineering complexities.

Overview

Having laid the foundation in the introduction, this report proceeds to elaborate on the distinct encounters and observations made in each department visited during the field training program. The upcoming segments will expound on the acquired insights from the dialysis department, followed by an examination of the maternity department and the central maintenance workshop. By thoroughly dissecting the operations, obstacles, and tangible implementations of medical engineering concepts within each department, this report aims to provide a comprehensive grasp of the training journey and its pertinence to the realm of medical engineering.

Dialysis Department

Demonstrating a deep dedication to elevating patient results and guaranteeing the smooth functioning of sophisticated medical apparatus, the medical engineering team assumes a crucial responsibility in upholding, fine-tuning, and rectifying the intricate machinery integral to the dialysis procedure.

Activities and Observations

- Clarification of the engineering and maintenance department's function within the dialysis unit, emphasizing its significance in upholding device safety and optimizing efficiency through preventive maintenance. Additionally, the department offers essential guidance to medical staff on optimal equipment utilization, thereby enhancing patient service quality. General discussion about the dialysis machine and the causes of chronic renal failure.
- Recognize internal parts of the Hemodialysis device and their respective functions.
- Acquire knowledge about disassembling and installing piston pumps.
- Conducting a comprehensive comparison of the distinct versions of the machines, namely Nipro, B-Braun, and Fresenius, with an aim to highlight their individual characteristics and features.
- Proficiently recognizing and differentiating both the internal and external components of the equipment, thereby gaining a comprehensive understanding of their intricate composition and functionality.
- Gaining practical insights into the process of rinsing through the utilization of hot citric, thereby acquainting oneself with the meticulous steps involved in this essential maintenance procedure.



Figure 1 Nipro dialysis machine in rinse mode

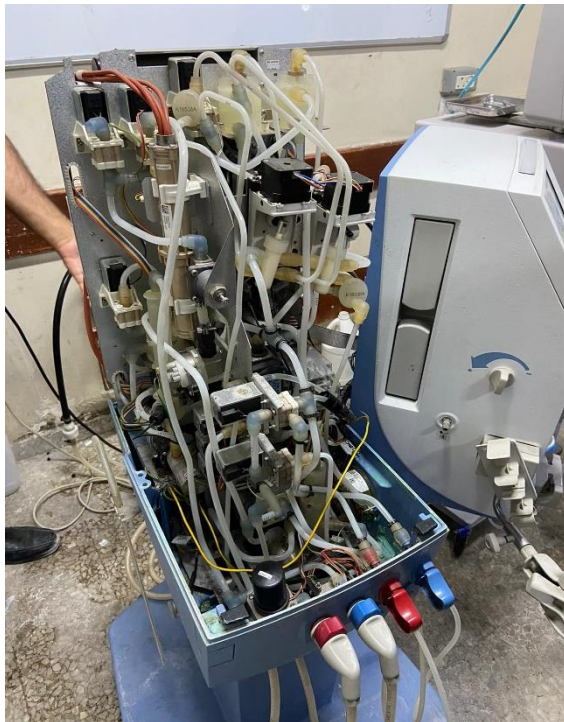


Figure 2 Inside the hemodialysis machine



Figure 3 Repairing process for conductivity error

Notes

- ✓ Dialysis machines need to be built in a way that allows the patient's tubing to be connected easily. All machines should have basic valves and sensors, but they vary in size, how they look, the kind of bicarbonate they use, and extra sensors and valves that make them work better and keep patients safe.
- ✓ The training was really helpful, even though it didn't last long. The engineers in the dialysis department were friendly and willing to answer any questions we had. They also explained how things are done in the department. Plus, our knowledge from classes came in handy.
- ✓ I have gained substantial knowledge about the department and its equipment. Moreover, I had a valuable hands-on experience during my time spent in the department, which I found to be quite beneficial.

Obstetrics and Gynecology department

In the Obstetrics and Gynecology department, the collaboration between medical engineering and patient care stands out. This part explores how medical engineers and healthcare staff work together to provide excellent care to mothers and newborns. The medical engineering team's careful adjustments, upkeep, and problem-solving for special tools like fetal monitors and advanced imaging machines are vital for safe deliveries and complete women's health services.

Activities and Observations

There was not much to learn in this section due to the lack of failed equipment there is no much equipment used in this department. Poor communication with the staff there was an issue also. The following are the activities in this department:

- Providing a basic overview of the Cardiotocography (CTG) device and how it operates.
- Monitoring the procedure for fixing a CTG device the experienced software related issues (the touch screen was disabled due to error in the software)
- Supervising the mending of a temperature sensor in a fetal monitor that had a wire cut.



Figure 4 CTG and its transducers

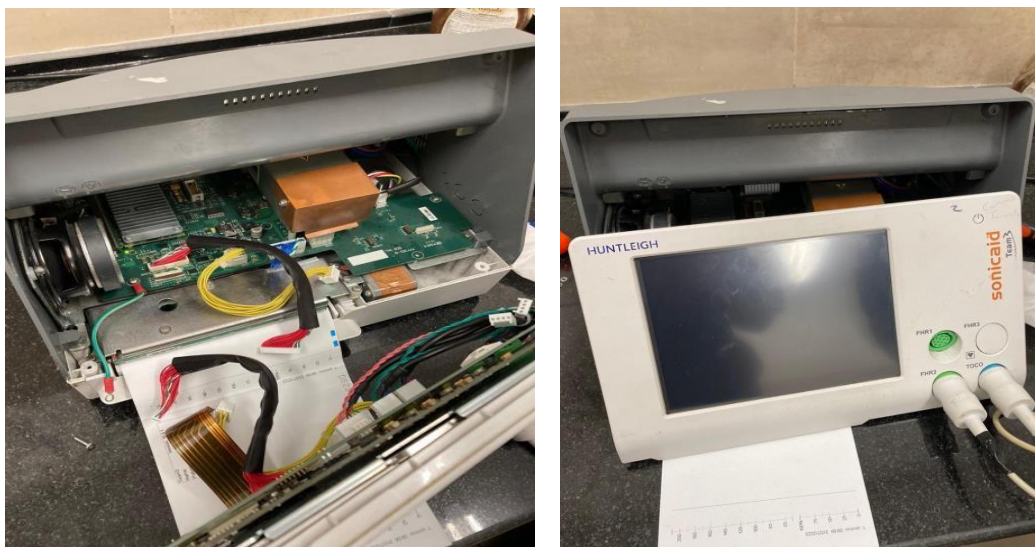


Figure 5 CTG device repairing process

Also, there was an error in the blood pressure monitor. There was a problem with the pump of the device as it was pumping and stopping due to a problem in the cuff. Changing the cuff with a new one was the solution of this problem.



Figure 6 Blood Pressure monitor with cuff problem

Central maintenance workshop

Imagine the workshop as the heart of the hospital. Experts there make sure medical machines work well. We'll explore what they do. Technicians and engineers use what they know to keep things working. By talking about what we learn here, we'll show how these skills help patients and the hospital.

Activities and Observations

• Engineers and Technicians Roles

Engineers and technicians in the central maintenance workshop play distinct yet complementary roles. Engineers, specialized in various fields like electronics and mechanics, focus on the design, analysis, and overall functioning of medical equipment. Technicians, on the other hand, possess hands-on skills in troubleshooting and repairing specific devices.

• Special Protocol for Device Repair

The workshop follows a structured protocol for device repair. It starts with medical staff handing over malfunctioning devices. Technicians then analyze the issues and determine if an engineer's expertise is needed. The device is sent to either an engineer or a specialized technician, who diagnose and fix the problem. Once repaired, the device returns to the respective department, ready for use.

• Electronics Discussion with Engineers

An interactive discussion takes place with electronics engineers in the workshop. This conversation delves into the intricacies of electronics, focusing on topics relevant to medical devices such as ECG machines.

• Infusion Pump Malfunction Demonstration

Workshop participants observe and even take part in resolving a technical glitch in an infusion pump. The device displays a "FLOW RATE error" message, and attendees witness the troubleshooting process.

- **X-Ray Device Explanation**

A specialized engineer provides a concise explanation of the functioning of the X-Ray device. This overview includes insights into its working principle and the roles of its various components.

- **Centrifuge Malfunction Demonstration**

Participants witness a demonstration involving a centrifuge machine with an error message. This provides a firsthand understanding of the diagnostic and repair procedures for such issues.



Figure 7 Centrifuge machine



Figure 8 Infusion pump

This collective experience in the central maintenance workshop sheds light on the crucial roles of engineers and technicians, the systematic repair protocol, insights into electronics, practical troubleshooting of different devices, and an overall appreciation for the intricate machinery that supports medical care.

Tours in the hospital departments

Extracorporeal Shock Wave Lithotripsy (ESWL) Unit

- Identifying the components of the unit and understanding the main idea of the work of the device in how to break up the stones in the kidneys.
- Dealing with a malfunction that was present on the screen (Water-Air-System Error).
- Also I have been seen an ultrasound imaging system with its transducers.



Figure 9 ESWL unit



Figure 10 UPS System for the ESWL unit



Figure 11 Ultrasound Imaging Machine



Figure 12 Ultrasound beam

ICU & OR

Take a tour of the Intensive Care Department to become familiar with its equipment, their operations, and how they're linked to patients. Delve into the inner workings of these devices and understand how they play a crucial role in patient care.



Figure 13 Portable X-Ray Imaging

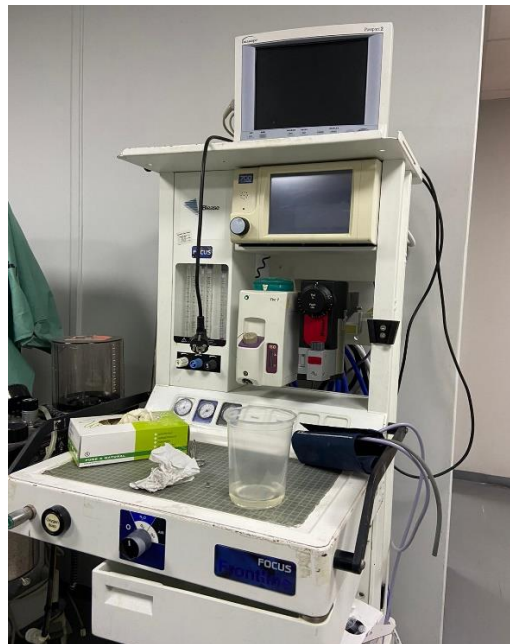


Figure 14 Anesthesia Machine

Conclusion

In a month-long, 60-hour course within Shifa Hospital's Medical Engineering Department, we delved into three crucial healthcare realms: dialysis, obstetrics and gynecology, and the central maintenance workshop. These hands-on experiences showcased the essential role of medical engineering in ensuring both precise patient care and smooth hospital operations.

In the dialysis and obstetrics departments, we witnessed the significant impact of accurately calibrated equipment on critical procedures, leading to better patient outcomes. The central maintenance workshop highlighted the indispensable contribution of technicians in maintaining the hospital's functionality.

Throughout the training period, we fine-tuned practical skills such as equipment maintenance, effective problem-solving, and interdisciplinary communication. This immersive experience deepened our understanding of the medical engineer's role and the profession's future in our country. It also allowed us to put academic knowledge into real-world practice, enriching our journey as aspiring medical engineers.